

Estimating Treatment Effects - Regression

February 25, 2025

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- Solution?

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$$\{Avg_n[Y^1|D=1] - Avg_n[Y^0|D=0]\} = \{E[Y^1|D=1] - E[Y^0|D=0]\}$$

Policy Evaluation - Randomization

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- So randomization is the ideal and the “gold standard” of research methodologies.
- But...in policy analysis it's often (though not always) impossible to randomize individuals in our sample into treatment?
- How can we attempt to identify causal effects of policy when randomization is infeasible?

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● Regression Analysis

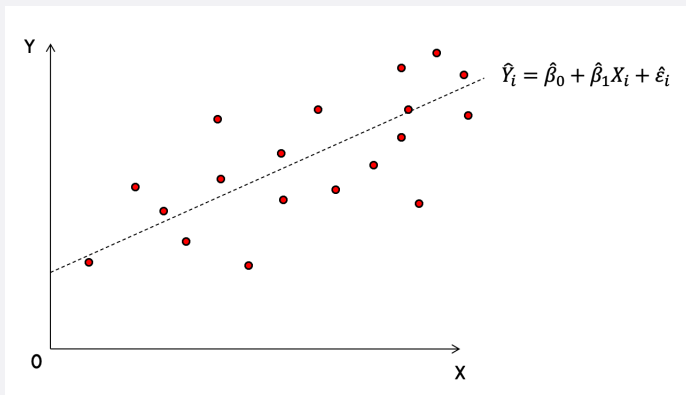
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- ▶ Even if D is not randomly assigned, we can still *hope* that $(Y^1, Y^0) \perp\!\!\!\perp D$.
- ▶ And that $ATE_{est} = ATE$.
- ▶ BUT THIS NOT “GUARANTEED” LIKE IT IS WITH RANDOMIZATION AND DEPENDS ON A STRONG (UNTESTABLE) ASSUMPTION.

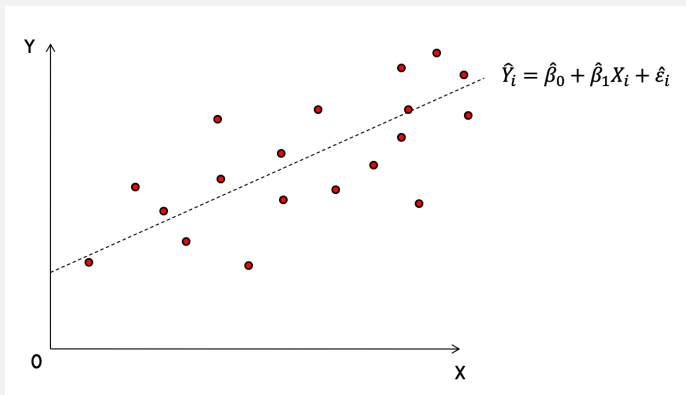
Policy Evaluation - Regression

- So how does regression actually work?



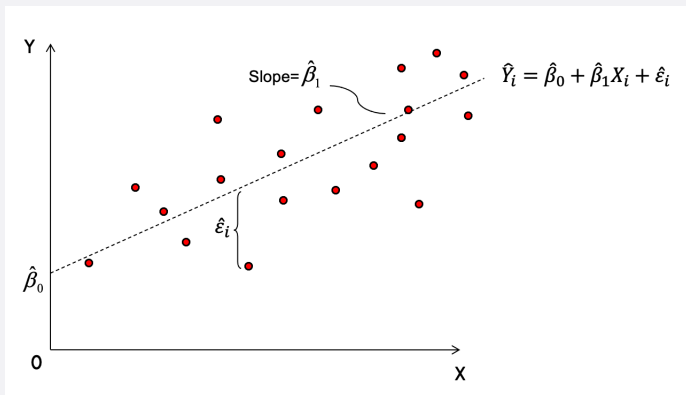
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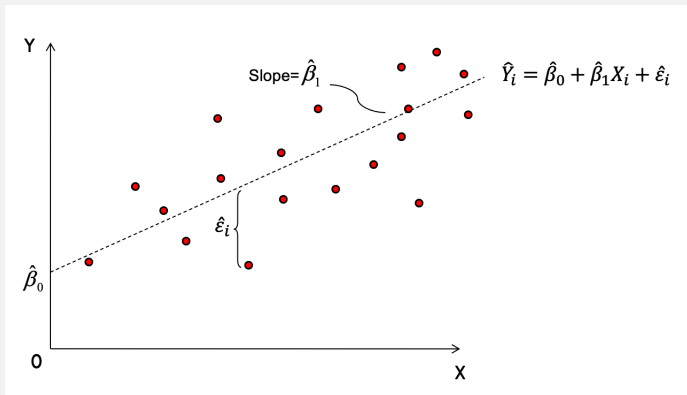


- Note that we have not randomized people into X in this case. We simply observe an individual's value of X in our sample.

Policy Evaluation - Regression



Policy Evaluation - Regression



- Most popular method for drawing the regression line involves minimizing the squared sum of the residuals. (Why the squared sum and not just the sum?)

- Suppose that we estimate a linear regression model of the following form: $y_{it} = \beta_0 + \beta_1 Edu_{it} + \varepsilon_{it}$

Where y is the number of times that individual i visits the emergency department in year t

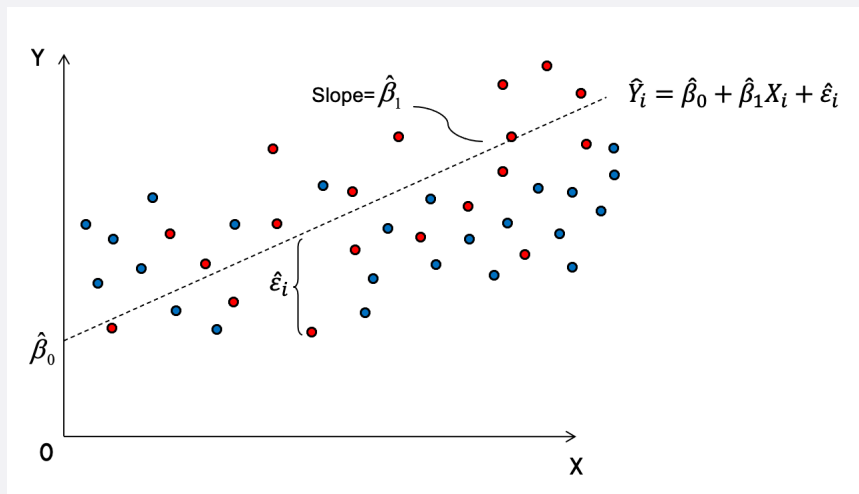
Edu is individual i 's education in years (e.g. a value of 12 represents a high school education, a value of 14 represents a college education, etc.).

Our coefficient estimates are as follows: $\hat{\beta}_0=4.0$ and $\hat{\beta}_1=-0.25$.

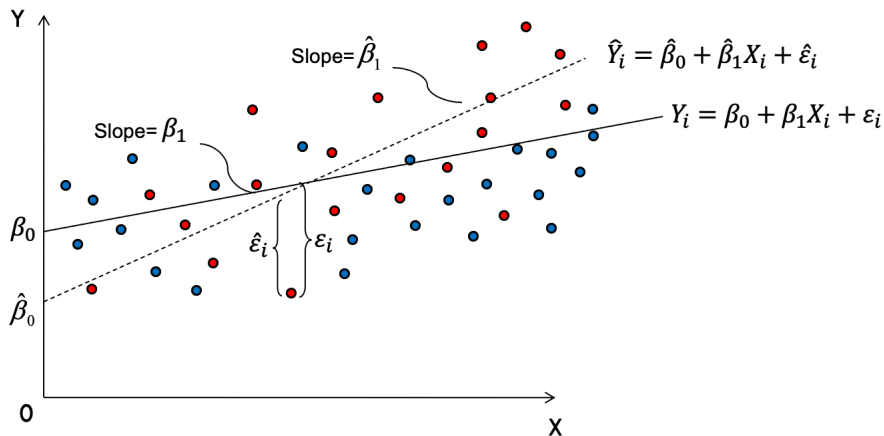
How do we interpret these coefficient estimates?

Policy Evaluation - Regression

- But suppose the *population* distribution is...



Policy Evaluation - Regression



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- ② Use the standard error to calculate a t-statistic and corresponding *p*-value
 - ▶ $C.I._{\hat{\beta}} = \hat{\beta} \pm t^* \times SE(\hat{\beta})$
- Suppose you estimate the regression model above and find that the coefficient estimate of β_1 has a *p*-value of 0.115 from a two-sided t-test. What does that mean and what would you conclude about the statistical significance of the relationship between education and emergency department use?

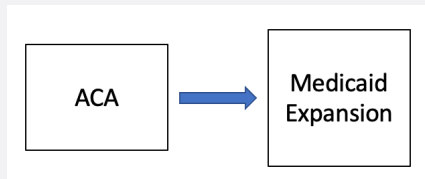
Policy Evaluation - Regression

- At this point, regression doesn't look all that different from randomization.

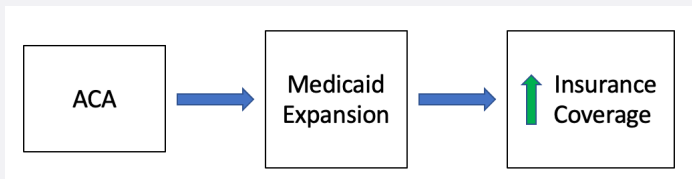
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- Where these methods begin to diverge is when we add additional covariates to our regression model to “control” for potential confounders.

- Medicaid Expansion and Mortality

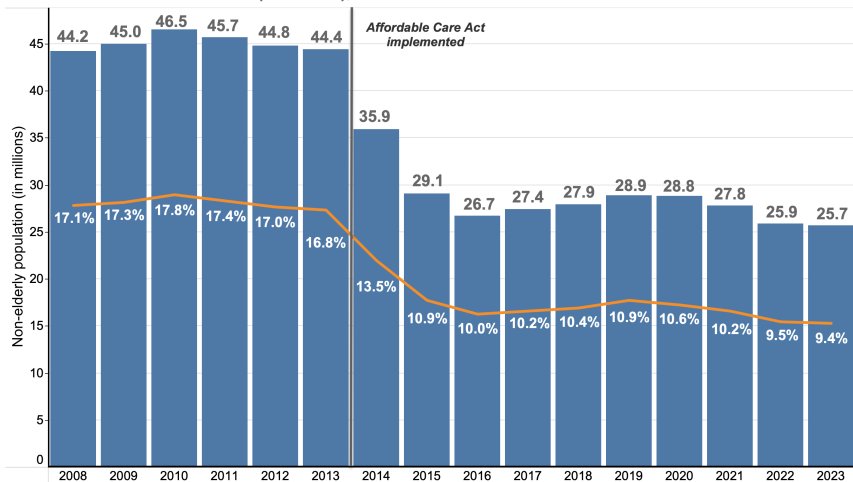


Policy Evaluation - Regression



UNINSURED IN THE UNITED STATES

NON-ELDERLY POPULATION (AGE 0-64), NUMBER AND RATE

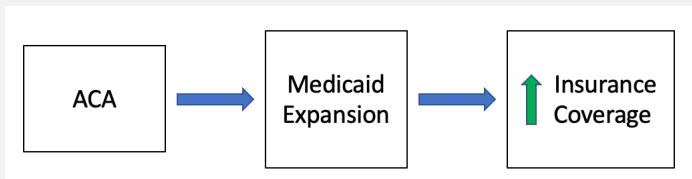


Source: Kaiser Family Foundation, analysis of 1-year ACS data

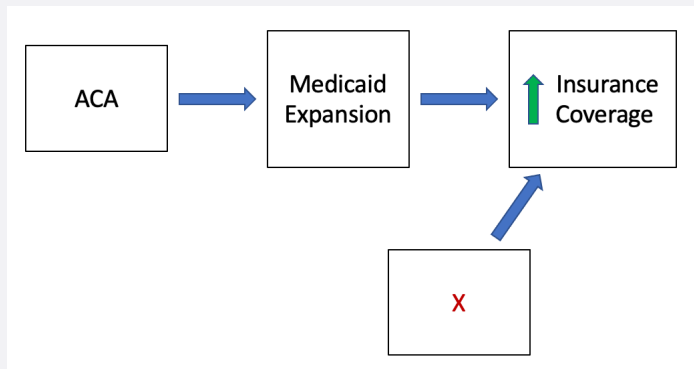
EconoFact

econofact.org

Policy Evaluation - Regression



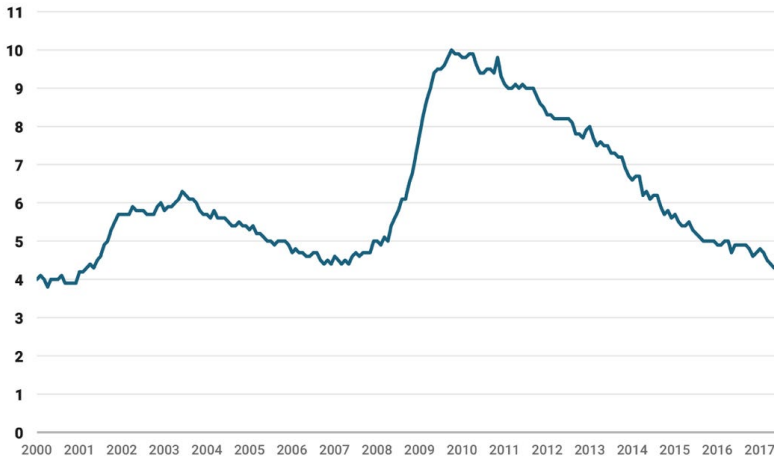
Policy Evaluation - Regression



MARKETS | CHART OF THE DAY

UNEMPLOYMENT RATE

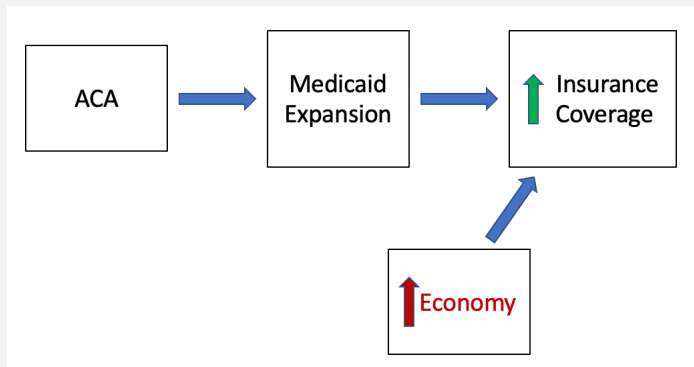
Percent of civilian labor force



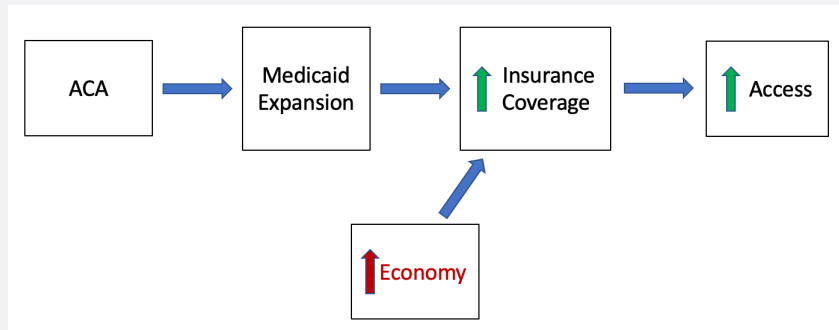
SOURCE: Bureau of Labor Statistics via FRED

BUSINESS INSIDER

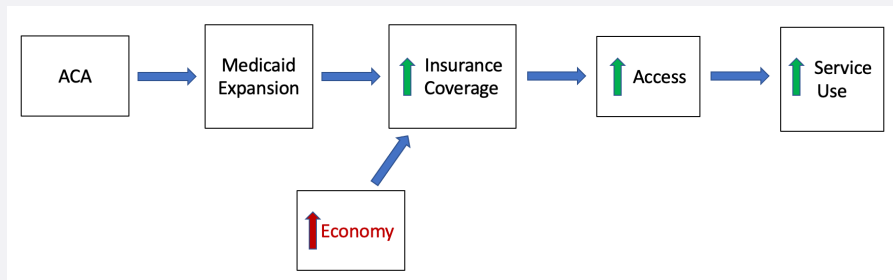
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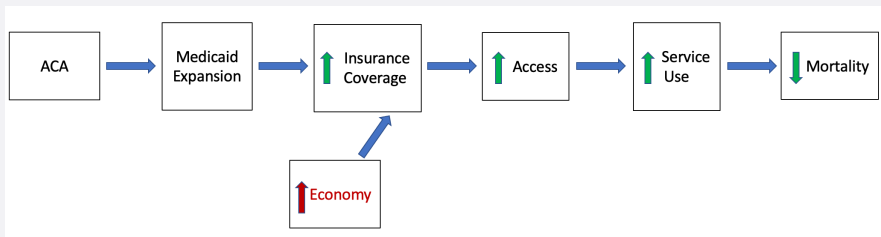
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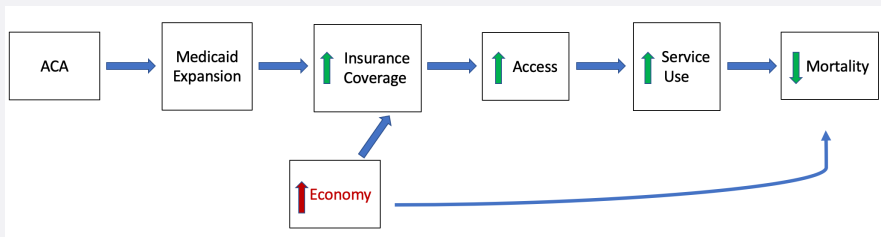
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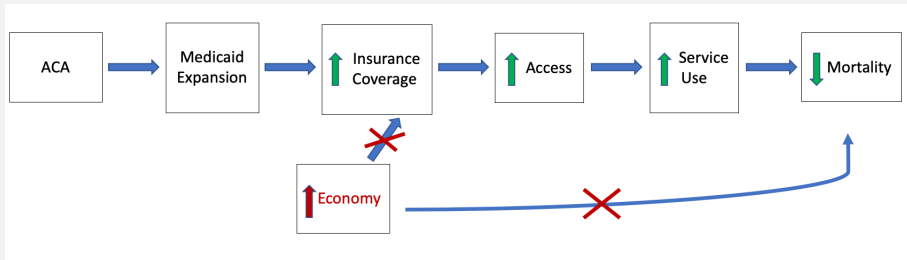
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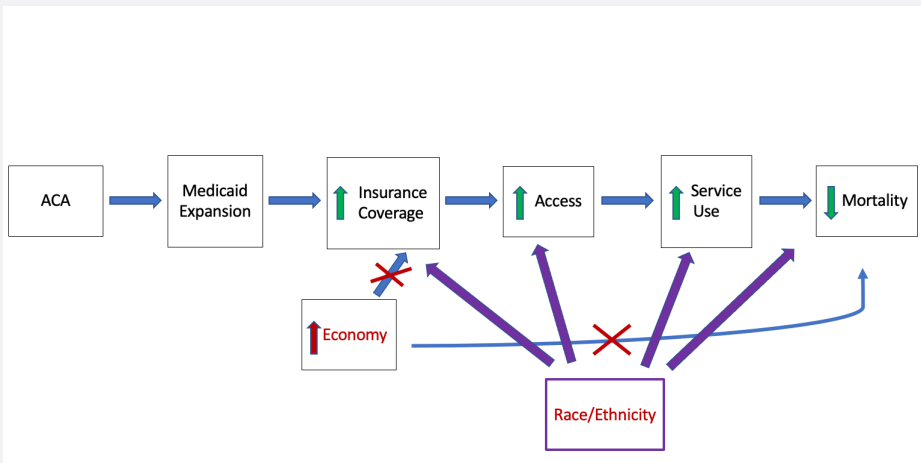
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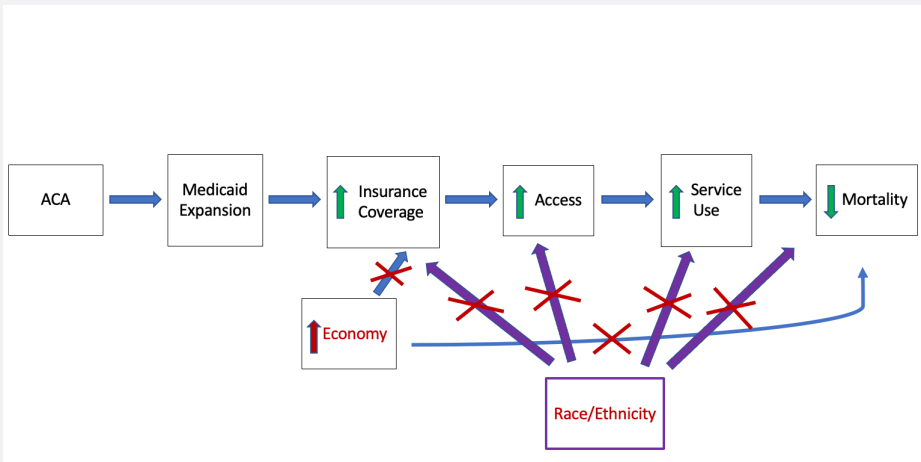
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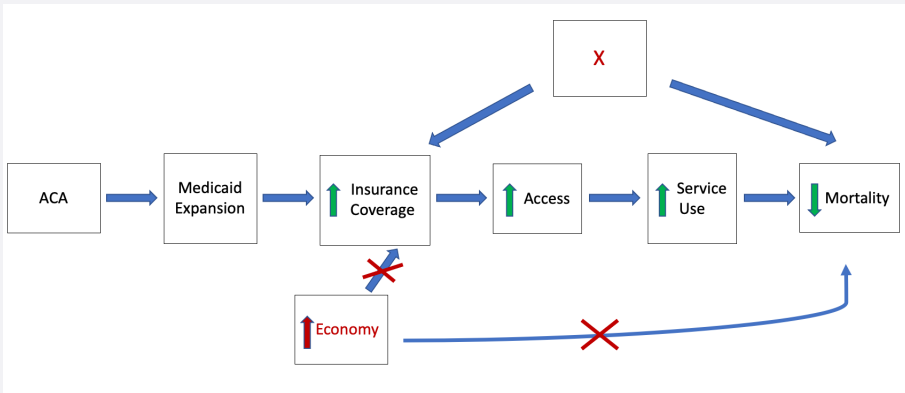
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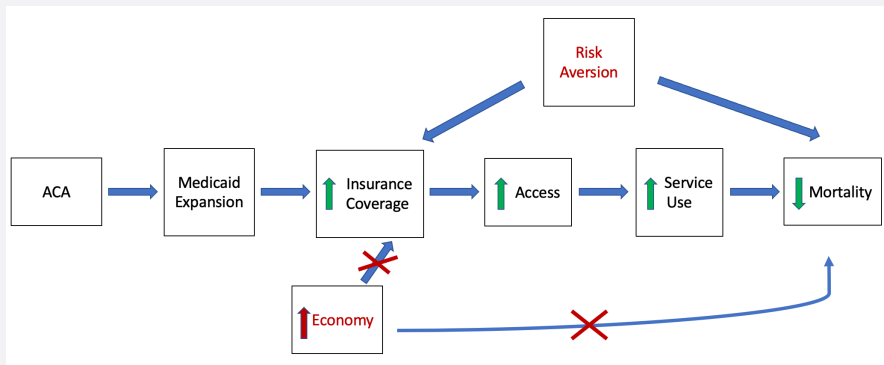
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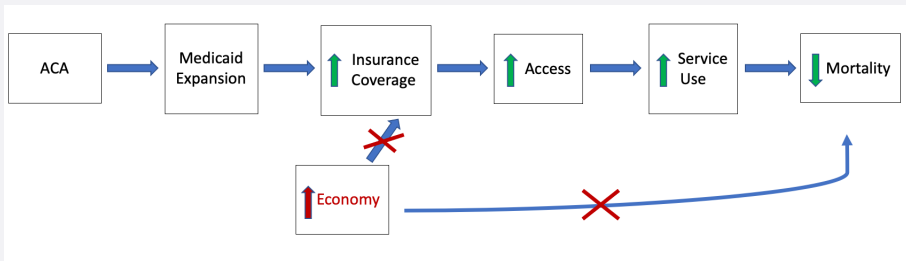
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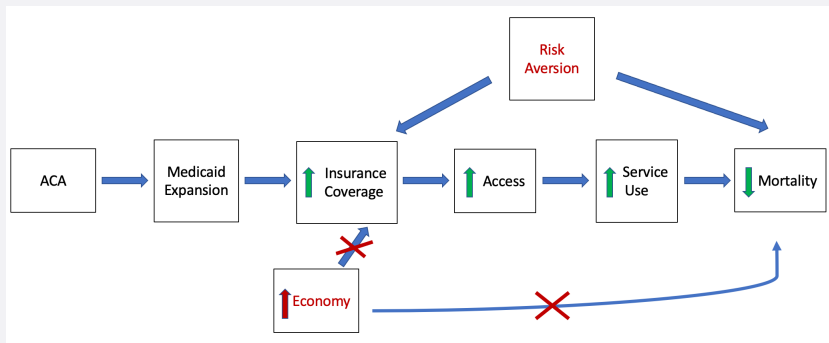
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- Sometimes we can determine which direction that bias is likely to take.
 - ▶ Suppose that those who are more risk averse are more likely to sign up for Medicaid coverage when eligible and are more likely to be in better health.
 - Then our estimate of the effect of Medicaid coverage on mortality will be biased *away from zero*.

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 - ▶ But omitted variable bias is a serious threat when exposure to treatment is non-random.
- So what do we do if randomization is infeasible and we're not confident that regression estimates will be unbiased?
 - ▶ Natural experiments and quasi-experimental research design.